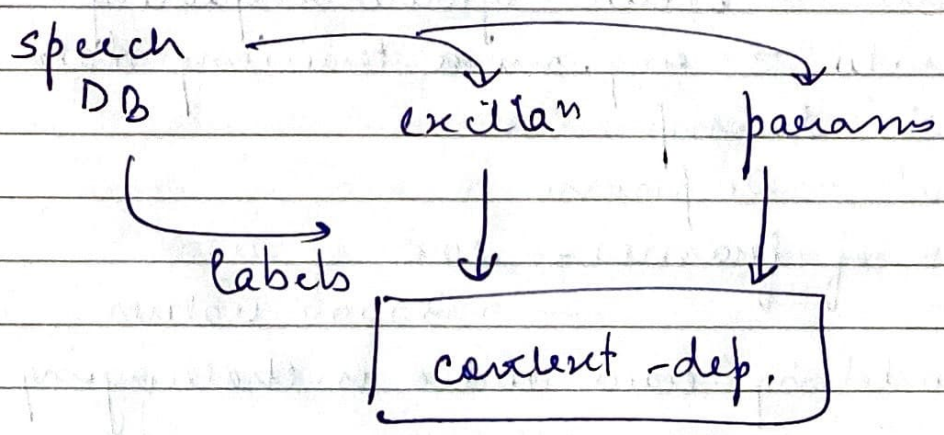


Week-12 Speech Tech

L1 Intro to TTS Synthesis

- speech - impt. mode of communication
 - TTS sys - automatic speech generation
 - ↳ corresponding to given text
 - ↳ audio - natural & intelligible speech
 - applicaⁿ - speech based, helps people with literacy difficulty, aid the visually challenged
 - discussion on brief history of TTS.
 - ↳ concatenative speech synthesis (railways)
 - Frameworks - USS, HTS, NN-, F2E
 - ↳ phase - Training & synthesis
- ↓
- < text, audio > pairs
- USS - unit selecⁿ synthesis
 - select & concatenate units from large speech database.
 - choice of units → phone (c or v), akshara (c⁺ v), diphone, syllable (c⁺ v c⁺), word
 - training → slice the speech DB at unit level, db should cover multiple realizaⁿ, fallback → word → syllable, akshara → phone
 - synthesis → appropriate units selected & concatenated, units chosen to min. target & concatenⁿ costs.

- adv. → natural sounding speech
- disadv. → req. very large db, discontinuities at concatenation pts.
- Lexicon / LTS / Grapheme-to-phoneme (G2P)
- alignment → waveform → spectrogram
- HTS (HMM-based) model
 - instead of storing actual waveform units, models of units
 - based on source-filter model of speech
 - extraction of features → source/MFCC
 - statistical parametric modelling is performed



- training - every phone is 3-state HMM
- synthesis → broken, corresponding HMMs are selected & concatenated, generate acoustic features which is fed to a vocoder.
- 2 models → duration & acoustic model.
- in HTS → context dependent pentaphone.
 - ↳ model for pentaphone (but monophone)
- challenges → some of combinations might not be present
 - ↳ binary trees, cluster the HMM states

- adv → low data is fine, small footprint size, fast syn, flexible (just HMM params)
- disadv → voice quality is poor.
- Neural network based
 - learn the mapping btw. linguistic & acoustic feature vectors using NN.
 - DNN, LSTM, RNN, BLSTM
 - train duraⁿ & acoustic models
 - synthesis ⇒ text → linguistic features → generate acoustic features → vocoder for speech constructⁿ.
 - adv. → better synthesis quality
 - disadv. → req. more training data
- TTS - decompresses
 - text - 20 phone
 - no. of frames - 200
 - 32000 values
- modeling raw audio is challenging.

L2 E2E Speech Synthesis - Auto-Regression Network

- TTS is decompressing
- WaveNet
 - raw waveform of audio directly modeled 1 sample at a time.
 - predictive distribuⁿ for audio sample

//_

conditioned on prev. samples
→ joint probability of waveform

$$x = x_1, \dots, x_T$$

$$p(x) = \prod_{t=1}^T p(x_t | x_1, \dots, x_{t-1})$$

→ auto-regressive model

→ conditional probability is modeled by a stack of convoluⁿ layers.

→ causal conv. → based on present & past.

→ training → predic^{ns} for all timesteps made in parallel → faster training due to no recurrent connec^{ns}

→ synthesis → sequential predicⁿ

→ to capture more context

→ inc. ↑ no. of layers

→ dilated convoluⁿ

→ network trained w/o text seq. ⇒ babbling

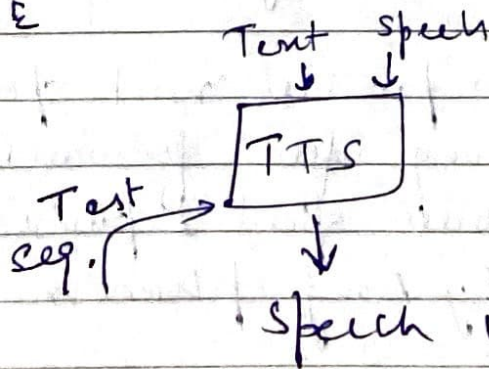
→ TTS → additionally condiⁿ on linguistic features⁽ⁿ⁾, or speaker identity.

→ adv. → good quality speech

→ disadv. → large dataset, huge computaⁿ

- earlier → parser, alignment, vocoder
↳ multi-step synthesis.

- E2E



no sep. modules.
allows rich
conditioning
single
model

- _/_/_
- has attenⁿ
- Tacotron-2 charac. seq.
 - ↓
 - seq2seq model with attenⁿ.
 - ↓ spectrogram frames
 - reconstructⁿ
 - ↓
 - speech waveform
- 2
slag¹
- but fastspeech & fastspeech 2 doesn't have attenⁿ
1. encoder & attenⁿ based decoder.
 2. vocoder.

- when to stop generaⁿ? stop token is used.
 ↳ probab. that generaⁿ is complete.

- issues with auto-regressive models -
 - not robust → repetitions & word skips
 - slow inference speed
 - lack of control over speed & prosody.

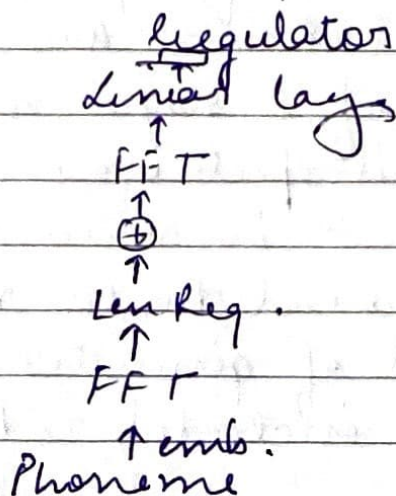
13

E2E - Non-Auto Regressive Networks

- FastSpeech → provide hard alignment btw. phonemes & mel spectrogram using a duratⁿ predictor.
- slow inference → parallel mel spectrogram generaⁿ using feed forward transformers
- lack of control over speed & prosody → adjust voice speed by Ding phoneme duratⁿ

→ fud-forward transforms / /

- archi. → FFT, N blocks on I/O sides, len.



- self attenⁿ → MHA to extract cross posⁿ info.
- 1D conv. layers (instead of dense networks in the transformer network)
- len. regulator → len. of phoneme seq. expanded to length of mel-spectrogram seq., speed up or slow down generated speech during synthesis.

$$H_{\text{mel}} = \text{LR} (H_{\text{pho}}, D, \alpha) \rightarrow [d_1, \dots, d_n]$$

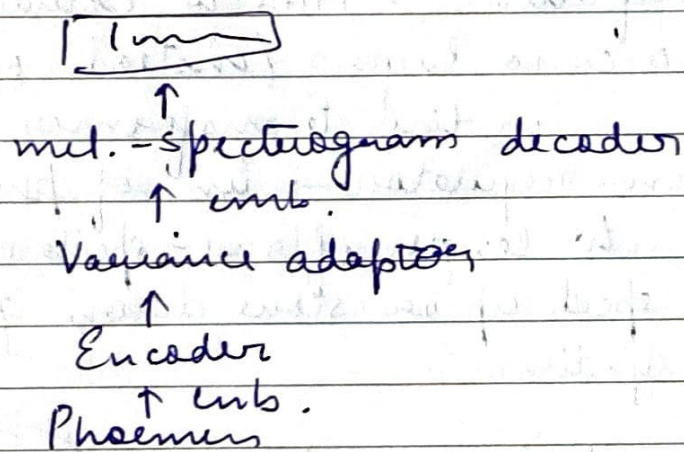
no. of frames

↳ HPT for speed control

$\alpha < 1$ fast speed
 $\alpha > 1$ slow speed

- duraⁿ predictor → 2 layers of 1D conv. network with layer norm; linear layer output predicts the phoneme duraⁿ. duraⁿ predictor is used only during the syn. phase.
- Knowledge distillaⁿ - duraⁿ info. for training data is by attenⁿ alignments that is generated from teacher network. Fast speech model is trained on text & mel-spectrograms.

- from fastspeech $\rightarrow$ fastspeech 2
 $\rightarrow$ teacher-student distillaⁿ has a complicated training process of 2-staged model. Info. loss of target mel spectrogram. Duraⁿ info. sometimes may not be accurate.
- Fastspeech 2 $\rightarrow$ ground truth mel-spectrogram are used instead of generated mels. Pitch & energy is also included to bring more reasonⁿ.



- training $\rightarrow$ alignment is obtained from (ext. aligner).
 $\rightarrow$ ground truth values of pitch & energy.
- synthesis $\rightarrow$ duraⁿ, pitch & energy predictor are used.
- Neural Vocoders
 - $\rightarrow$ WaveNet
 - $\rightarrow$ WaveGlow
 - $\rightarrow$ GAN-based $\rightarrow$ MelGAN, etc.

- E2E TTS

→ remove mel-spectrogram as intermediate representation.

→ waveform contains more info. Eg. VITS, F2

→ we can include additional embeddings.

like speaker, lang., global style tokens

→ adv. → high quality speech, easy to train

→ disadv. → huge data, computationally expensive

- Evaluation of TTS Sys.

→ obj. → mel cepstral distorⁿ scores

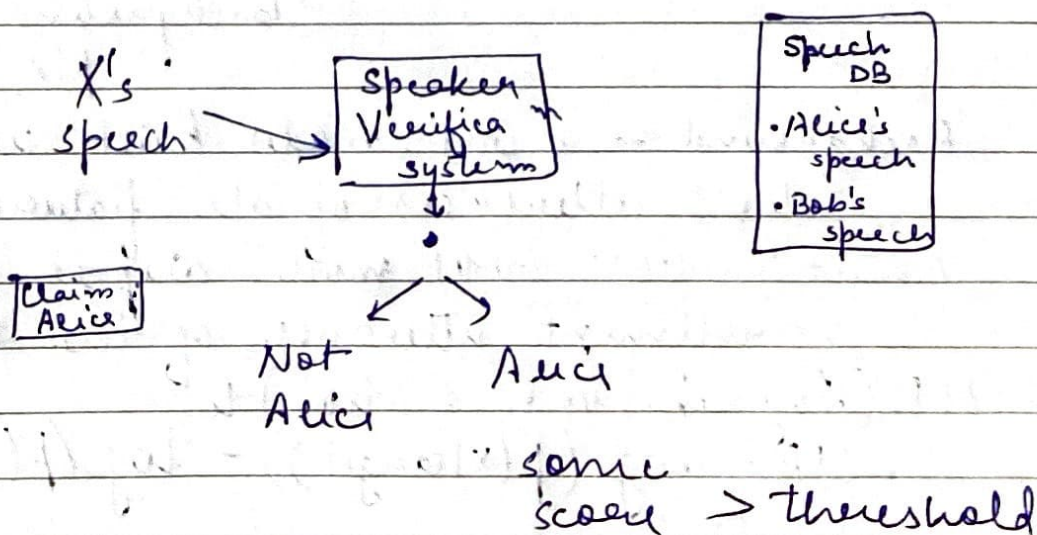
→ subj. → mean opinion score (MOS)

→ degraded MOS (DMOS)

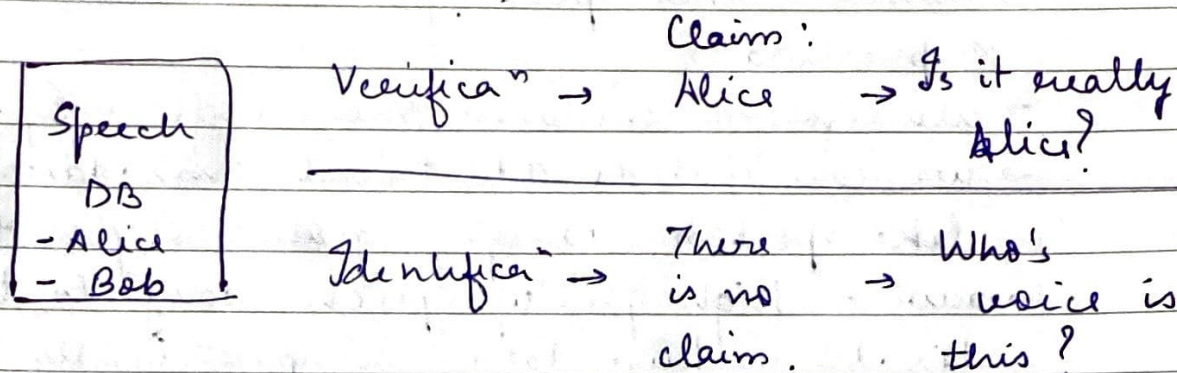
→ pairwise comparison (PC) test

- future research areas → multilingual, prosody, voice conversion, conversational speech

Intro to Speaker Verification



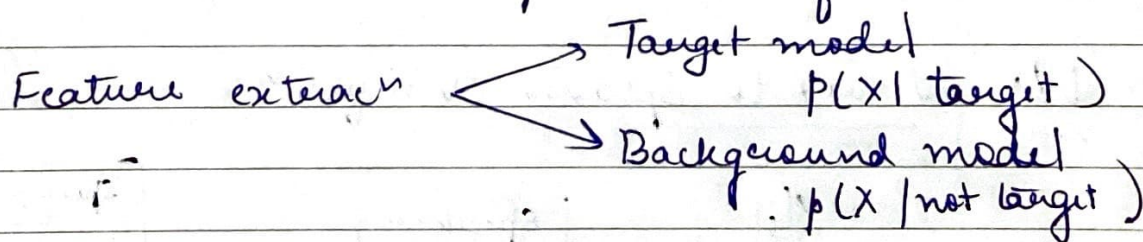
- What is speaker verification vs identification



- Types of speaker verification
- Text-dependant
 - Text-independent (TI)

TD	TI
- Sys. knows the text spoken	- Sys. does not know the text spoken
- knowledge of text spoken can improve performance	- more flexible but difficult problem.

- UBM-GMM-based speaker verification



Background → a GMM model built with features from utterances of all features.

Target → UBM-GMM model adapted to the enrollment utterance of the speaker.

- LLR (Log likelihood ratio) test
- $$LLR = \log(p(X | \text{target})) - \log(p(X | \text{not target}))$$

$LLR > \text{threshold} \Rightarrow \text{accept}$

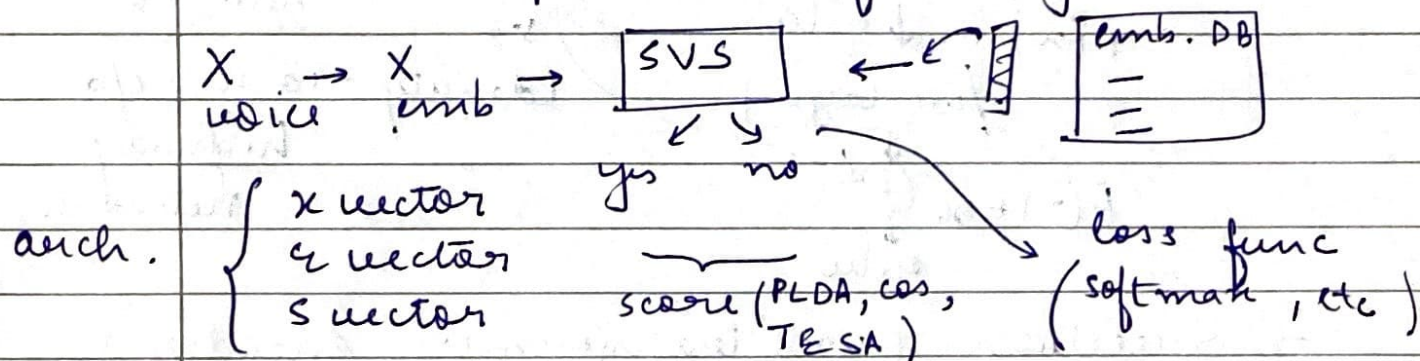
- When the means of GMM are stacked, the resultant vector is cld a supervector. If there are k components of a J -dimensional GMM, the resultant supervector is of dim $k \times J$.
- I -vectors $\rightarrow$

$\rightarrow$ speaker & channel dep. supervector

$$M = m + Tw \rightarrow \text{the } i\text{-vector}$$

low rank Total variability matrix. $\rightarrow$ speaker indep. supervector from UBM-GMM.

- NN-based speaker verification sys.



I $\rightarrow$ train a NN for speaker classification with large no. of speakers

II $\rightarrow$ get the speaker embeddings

III $\rightarrow$ score the embedding of a trial pair

LS Speaker Classification Architectures

- X -vectors

I $\rightarrow$ train a TDNN based NN

$\rightarrow$ TDNN blocks are made of conv 1D blocks, activation funcⁿ, batch norm.

TDNN $\rightarrow$ Time-delayed NN

$\rightarrow$ conv ID (- - -)

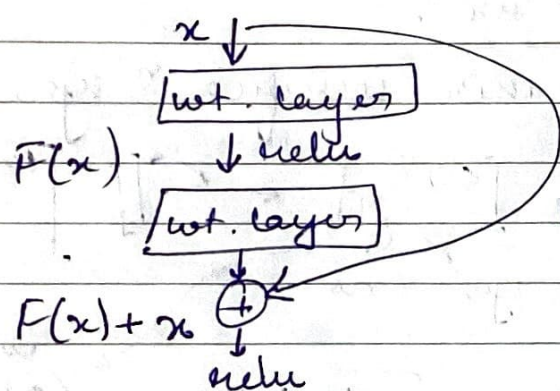
$\rightarrow$ speaker embeddings are extracted from the affine layer of FFNN-1

- R-vectors

$\rightarrow$ resNet blocks are made up of conv 2D blocks

$\rightarrow$ ResNet 34 arch. $\rightarrow$ eg.

$\rightarrow$ ResNet



$\rightarrow$ very helpful in training deep networks

$\times$ identity $\rightarrow$ also c/d highway networks

- S-vectors (uses transformer encoder)

$\rightarrow$ speaker embeddings are extracted from the affine layer of FFNN-3.

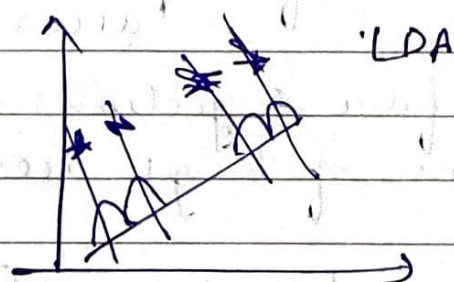
L6 Scoring & Evaluation Techniques

- PLDA (Probabilistic Linear Discriminant Analysis)

Training $\Rightarrow$ extract $\rightarrow$ LDA $\rightarrow$ PLDA embed.

Scoring $\rightarrow$ Trial Pairs $\rightarrow$ Trained PLDA $\rightarrow$ Scores

- A trial pair is a set of 2 occurrences. 2 occurrences can be from same speaker (target access) or diff. speakers (imposter access).



- Cosine Scoring

Trial pairs $\rightarrow$ extract embed $\rightarrow$ cosine similarity $\rightarrow$ Scores

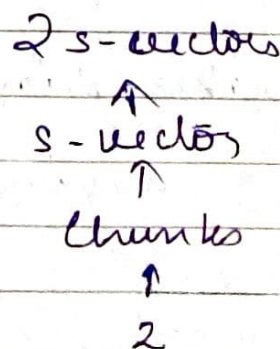
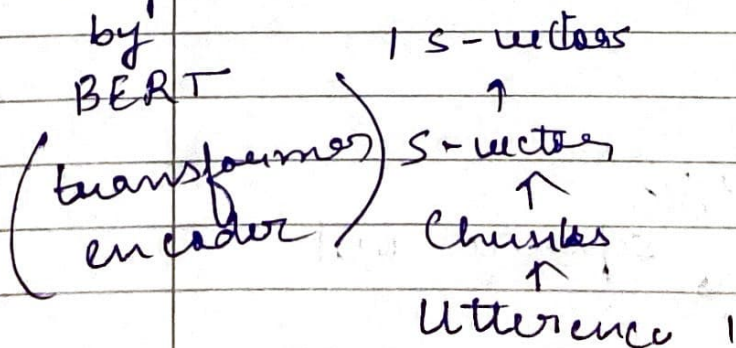
$$\text{simi}(p, q) = \cos \theta = \frac{p \cdot q}{\|p\| \|q\|}$$

- TESA (Transformer Encoder Speaker authentication)

BERT loss = NSP + MLM loss

$\hookrightarrow$ self-supervised learning

inspired by BERT



- trial pair scores
 - low → imposter access
 - high → target access

$$R_{FA} = \frac{\text{no. of false acceptance}}{\text{actual no. of imposter access}}$$

$$R_{FR} = \frac{\text{no. of false rejections}}{\text{actual no. of target accesses}}$$

- EER & Min DCF are based on R_{FA} & R_{FR}
- DET curves are obtained by varying the threshold & calculates R_{FA} & R_{FR} .
- EER
(equal error rate) $\Rightarrow R_{FA} = R_{FR}$
- DCF

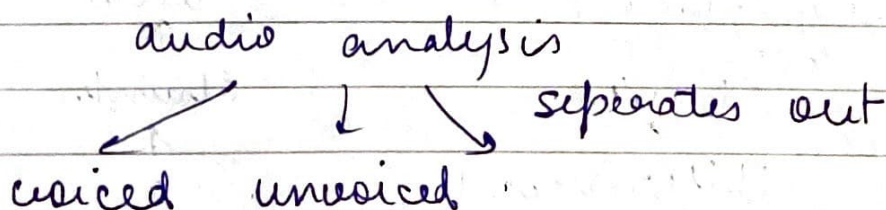
$C_{FR} \rightarrow$ cost of false reject
 $C_{FA} \rightarrow$ cost of false acceptance

DCF = $C_{FR} \times R_{FR} + C_{FA} \times R_{FA}$

↳ minimize this

← "detic" cost func"

L7 Speaker diarization



- 11
- speaker diarizaⁿ with x-vectors
speech activity detecⁿ
(SAD)



segmentaⁿ



x-vector extracⁿ



pairwise scoring



agglomerative hierarchical clustering

- SAD → mostly energy based, short term energy is computed
- segmentaⁿ → after SAD, signal is built into segments of fixed window len., each segment is 1.5s long, usually a overlap is allowed.
- then x-vectors are extracted
- the pairs are generated, a pairwise scoring is performed.
- finally $K = ?$ is defined, based on that agglomerative hierarchical clustering is done.
- the problem is clustering based techniques can't handle speaker overlaps. They are not trained to directly diarizaⁿ (min.) errors.
- E2E → can handle overlaps, directly trained to minimize errors, uses a loss funcⁿ named permutation free loss.

//_

Linear layer + sigmoid

↑
Transformer
↑ Encoder

Input features

- permutation free loss → takes care of both the ground truths
 $N_{\text{speakers}} \rightarrow N_p$